0093

National University of Computer and Emerging Sciences

Chiniot-Faisalabad Campus

Programming Fundamentals CS 1002

Final Term Exam

Date: December 16th 2024

Course Instructor(s)

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Total Time : 150 Min

Total Marks: 60

Total Questions: 4

Roll No _____ Section _____ Student Signature _____

Subjective

Subjective part is required to be distributed after collecting objective part.

CLO #: 4 Design basic problems of the real world through small/medium size programs given as course projects.

Q1	**Use of any kind of loops is not allowed for the following question.**	15

Write a C++ program to simulate a barcode reader. The barcode is an 8-digit number containing the following information:

- The **leftmost digit** represents the origin of the item:
  - 1: Local
  - 2: Imported
- The **rightmost two digits** represent the type of item:
  - 10: Refrigerated
  - 11: Frozen
  - 12: Others
- The **next (middle) five digits** represent the price of the item in rupees.

For frozen items, a fixed freezing charge of PKR 100 is added.

Local items have no tax, but imported items are subject to a 42% tax on the base price, excluding freezing charges.

Your task is to read the barcode, determine the item category, and calculate the total price after applying the relevant charges, taxes and display relevant item information. Take an integer number from console for testing purposes.

Handle wrong inputs as well.

Sr#	Sample Input	Sample Output
1	20043211	Category: Frozen Base Price: PKR 432 Import Tax: 42% of PKR 432: 181.44 Freezing Cost: PKR 100 Total Price: PKR 713.44
2	11043212	Category: Others Base Price: PKR 10432 Total Price: PKR 10432
3	1104321	Incorrect barcode!
4	41043212	Incorrect barcode!

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Q2	**CLO 3 Justify problem solving techniques and analytical thinking by identifying the concepts and properties of algorithms**	10+5

a) Implement a program that takes a 2D array (matrix) of integers and rotates it 90 degrees clockwise. Your program should not create a new array and should modify the input array in place.

Requirements:

- Use nested loops to manipulate the array.
- Include error handling for cases where the array is not square (e.g., return an error message).

Example:

matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]
Output: [[7, 4, 1], [8, 5, 2], [9, 6, 3]]

b) Consider the array given above, describe how to perform the following operations on a 2D array without using built-in libraries:
Find the maximum value in each column and store the results in another 1D array

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Q3	**CLO 4: Design basic problems of the real world through small/medium size programs given as course projects.**	15

a. Write down two overloaded functions named as SubAndDivide. One of these will have two integers and other will have two double parameters. Which ever function is called from any where it will print its prototype and will also print the number of times this function has been called. The functions will return the Difference and Division of the two numbers passed to it as well.

b. Write down one function that can take either one parameter, two parameters or three parameters and will return the sum of the numbers passed.

c. A 2D array, Picture, contains data representing a bitmap image. The second dimension is declared as a global named constant DIM2. Each element of the array represents one pixel of the image. The image is grey-scale encoded where the value of each pixel ranges from 0 (representing black) to 255 (representing white) with intermediate values representing different levels of grey. A graphics program needs a function Flip(), to flip (horizontal) the image. An example of an image before and after the function is:

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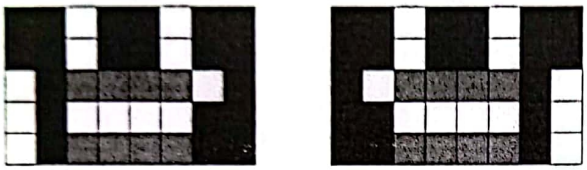
Before flip **After flip**



The values contained in the 2D array before the flip are as follows:

Data values

80	80	255	80	80	255	80	80
80	80	255	80	80	255	80	80
255	80	120	120	120	120	255	80
255	80	255	255	255	255	80	80
255	80	120	120	120	120	80	80

Write down C++ functions that contain the following functionality:

1. To display elements of any 2D array
2. Flip any 2D array passed to it as shown above

Note: In the first function the passed array must be set to be read only. **The array must be of char type but the output must be the integer values.**

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Q4	**CLO 4: Design basic problems of the real world through small/medium size programs given as course projects.**	15

Write a program, which reads a file, named "Students.txt", calculate the marks (3 courses) of each student from the file, and write the total marks of each students at the end of the same text file.
Include all required header files and handle error-checking mechanism if file does not exist or cannot be open.
Example: file structure is given below

```
Waseem      70      50      30
Ali         60      53      96
Jabbar      56      96      85
Ahmad       78      98      53
Uzair       79      89      84

Waseem      150
Ali         209
Jabbar      237
..
..
```